
Desarrollo de una herramienta en Python para el postprocesado de modelos térmicos transitorios realizados con ESATAN-TMS

TRABAJO DE FIN DE GRADO
ESPECIALIDAD: PROPULSIÓN AEROESPACIAL

Autor: Asier Fiel Mariblanca

Tutor: David González Bárcena

MADRID, SEPTIEMBRE 2026

Contents

1. Abstract	1
2. Introduction	3
2.1. Objectives	3
3. State of Art	5
3.1. Thermal Tools in the Aerospace Industry	5
3.2. Post-Processing Capabilities in ESATAN-TMS	5
3.3. Custom Interfaces: JPL Example	5
3.4. Practical Application: FalconSat-2	5
3.5. Motivation for the Present Work	5
4. Theoretical Background	7
4.1. Thermal Control in Engineering	7
4.2. Space Thermal Environment	7
4.3. Heat Transfer Mechanisms	8
4.3.1. Conduction Heat Transfer	8
4.3.2. Radiation Heat Transfer	10
4.4. Mathematical Modeling of Thermal Systems	10
4.4.1. Disadvantages of the Finite Element Method (FEM)	10
4.4.2. Lumped Parameter Method (LPM)	10
4.5. Software and Advanced Methods for Thermal Simulation	12
4.6. Analytical Capabilities of ESATAN-TMS	12
5. Development of the Thermal Model in ESATAN-TMS and Data Export/Analysis	13
5.1. Subsystem Classification	13
5.2. Materials and Properties	13
5.2.1. Thermo-optical properties	13

5.3. Steady-state radiative case	13
5.4. Boundary Conditions and Loads Definition	13
5.5. Model Verification: Group-Level Energy Balances	13
5.6. Data provided by the TMD	13
5.6.1. Use in the Python post-processing	13
6. Design and Implementation of the Post-Processing Code	15
6.1. Code Structure and Architecture	15
6.1.1. Motivation for a Modular Architecture	15
6.1.2. Files overview	15
6.2. Software Stack and Dependencies	15
6.3. Filtering and Grouping Methods	15
6.4. Figure Conventions and Representation Criteria	15
6.5. Scope of Examples and Cryogenic-Flow Validation	15
7. Results and Visualisations from the Post-Processing Workflow	17
7.1. Filtering Results	17
7.2. Sankey Diagram	17
7.3. Network Graph Views (NX)	17
7.4. Stacked-Bar Balances	17
7.5. Cryogenic-Flow Views (ARIEL)	17
8. Conclusions and Future Work	19
8.1. Conclusions	19
8.2. Future Work	19
References	21
Anexos	23

List of Figures

4.1. One-dimensional heat conduction accross the wall, with thickness Δx ,whose surfaces are at temperatures T1 and T2 ($T1 > T2$), respectively [2]	9
4.2. Simplified heat transfer system and its corresponding thermal network [8]	11

List of Tables

Acronyms

European Space Agency Thermal Analysis Tool (ESATAN-TMS)	1
Lumped Parameter Method (LPM)	10
Thermal Control System (TCS)	7

1. Abstract

This Bachelor's Thesis focuses on implementing post-processing tools for the thermal analysis of space systems, leveraging results obtained from the ESATAN-TMS software.

Building upon a previous study limited to a steady-state radiative case, the main objective is to evaluate the dynamic thermal behavior of the system by applying a fully transient model.

Since ESATAN-TMS presents limitations regarding integrated data visualization, this project develops custom post-processing routines in Python, leveraging libraries such as NumPy, Pandas, and Matplotlib. These tools automate the extraction of nodal data to generate advanced graphical representations, including temperature evolution curves and transient heat fluxes over time. Ultimately, this methodology optimizes the interpretation of complex thermal results, facilitating the comparison of different scenarios and ensuring the system's performance under realistic orbital thermal cycles.

2. Introduction

2.1. Objectives

3. State of Art

3.1. Thermal Tools in the Aerospace Industry

3.2. Post-Processing Capabilities in ESATAN-TMS

3.3. Custom Interfaces: JPL Example

3.4. Practical Application: FalconSat-2

3.5. Motivation for the Present Work

4. Theoretical Background

4.1. Thermal Control in Engineering

In engineering applications, thermal control is decisive to optimize the performance of the system and its components, while keeping them within temperature limits. In some systems and elements, such as engines, electric components or aerospace systems, the implementation of a Thermal Control System (TCS) is critical because thermal variations can produce a negative impact on their performance and structural safety [1].

Methods such as convection, conduction and radiation can be implemented on Earth's surface to study heat transfer phenomena. However, the lack of an atmosphere in space, limits heat transfer mechanisms to conduction in solids and thermal radiation into space [2].

The primary objective of a Spacecraft TCS is to ensure that all onboard components operate within their specified temperature ranges during all mission phases, while minimizing resource consumption (mass and power) [3]. This is achieved through a careful balance of internal heat dissipation and external environmental fluxes, utilizing both passive and active thermal control technologies [1].

Those aspects are critical because, unlike terrestrial applications, the space environment is characterized by extreme conditions, such as vacuum, a background temperature of 3K, and rapid, intense thermal cycling due to exposure to sunlight and Earth's shadow [2,3].

Failure to comply with these thermal limits can lead to equipment malfunctions, accelerated material degradation, reduced reliability, structural damage from thermal cycling, or even catastrophic failure.

The objective of thermal control is to prevent those situations, ensuring operational reliability and enhancing the efficiency of the system [4]. Some key requirements for the TCS include [5]:

- Temperature Ranges: maintaining specific operational and survival temperature ranges for components: electronics, batteries or infrared detectors
- Thermal Gradients and Stability: controlling temperature gradients and stability is essential for high-precision systems such as cameras, optical benches, and CCD sensors to avoid structural distortions and measurement errors

4.2. Space Thermal Environment

Spacecraft thermal control is a process of energy management in which environmental heating plays a major role and therefore, has a major impact in the spacecraft design. There is no convection in

space due to the absence of an atmosphere, being radiation the only possible method to dissipate heat in space.

Nevertheless, heat conduction between the structure and the components of the spacecraft still occurs. [1]

4.3. Heat Transfer Mechanisms

Heat can be defined as the form of energy that can be transferred from one system to another as a result of temperature difference. The science that deals with the determination of the rates of such energy transfers is heat transfer [4]. Conduction, convection and radiation are the fundamental mechanisms of heat transfer. However, as it is mentioned earlier, there is no convection in space systems, leaving conduction and radiation as the predominant mechanisms. [1]

4.3.1. Conduction Heat Transfer

Heat transfer conduction is the mechanism by which thermal energy is transmitted through solids or between solids in contact due to the presence of a temperature gradient, without any macroscopic motion of the substance. This process is governed by microscopic interactions between particles and, in metallic materials, the transport of free electrons [4].

From a macroscopic perspective, heat conduction is described by Fourier's law:

$$q = -kA \frac{dT}{dx} \quad (4.1)$$

where:

- q is the heat transfer rate (W)
- k is the thermal conductivity (W/mK)
- A is the cross-sectional area (m²)
- $\frac{dT}{dx}$ is the temperature gradient (K/m).

The negative sign in the temperature gradient indicates that heat flows from regions of higher temperature to lower temperature [2, 4].

Under steady-state conditions and simple geometries, Fourier's law can be integrated to obtain analytical expressions for heat transfer. For instance, in a plane wall with constant thermal conductivity, the heat flux is directly proportional to the temperature difference between the boundaries and inversely proportional to the wall thickness [4].

Figure 4.1 illustrates one-dimensional heat conduction through a flat wall, consistent with the formulation of Fourier's law in Equation 4.1.

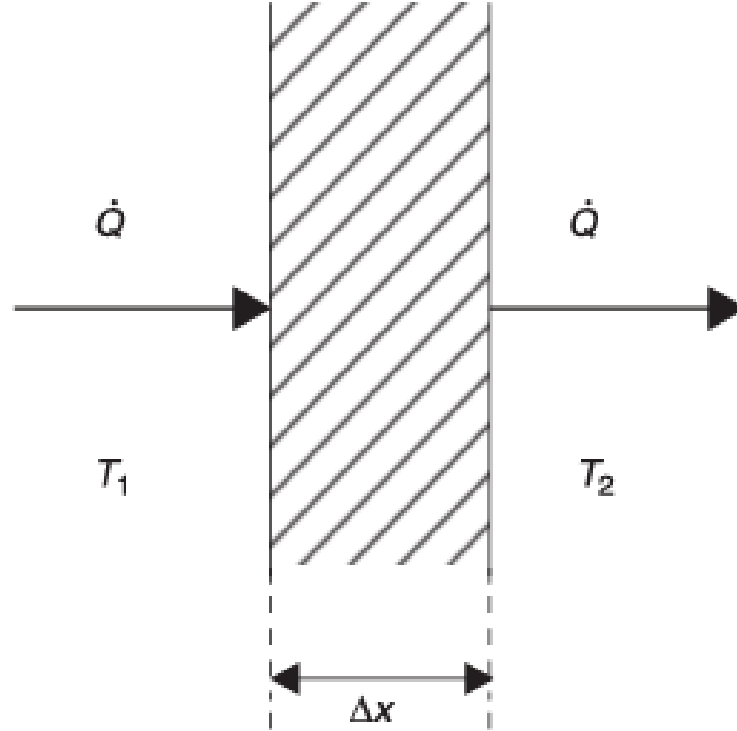


Figure 4.1: One-dimensional heat conduction across the wall, with thickness Δx , whose surfaces are at temperatures T_1 and T_2 ($T_1 > T_2$), respectively [2]

For more general cases involving spatial and temporal variations, the heat conduction process is governed by the heat diffusion equation, expressed in cartesian coordinates as:

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + \dot{q} \quad (4.2)$$

where ρ is the density (kg/m^3), c_p is the specific heat (J/kgK), and $\dot{q}$ represents internal heat generation per unit volume (W/m^3). This equation forms the basis for thermal modelling in both analytical and numerical approaches, including finite difference and nodal methods [4, 6, 7].

An additional useful concept in conduction analysis is thermal resistance, which simplifies the study of heat transfer in complex systems. For a homogeneous layer, the conductive thermal resistance is defined as:

$$R_{cond} = \frac{L}{kA} \quad (4.3)$$

where L is the thickness of the material.

In multilayer structures, individual resistances can be combined in series or parallel, allowing the overall heat transfer to be evaluated using an approach analogous to electrical circuits [4].

In the context of spacecraft thermal control, conduction plays a fundamental role in the internal redistribution of heat. Conductive paths are essential to transport heat generated by onboard components towards external radiative surfaces. Therefore, the design of conductive interfaces, material selection, and contact conductance are critical factors in ensuring proper thermal management [1, 2, 7].

Furthermore, in thermal modelling tools such as ESATAN, conduction is typically represented through thermal conductances connecting discrete nodes within a network. These conductances depend on material properties and geometry and are used to solve the energy balance equations governing the thermal behaviour of the system [6].

4.3.2. Radiation Heat Transfer

4.4. Mathematical Modeling of Thermal Systems

Mathematical modeling is essential to predict the thermal behavior of space systems and design effective thermal control strategies.

4.4.1. Disadvantages of the Finite Element Method (FEM)

4.4.2. Lumped Parameter Method (LPM)

The Lumped Parameter Method (LPM) is a discretization technique that simplifies a continuous physical system into a finite number of discrete elements called nodes. The fundamental assumption is that each node is isothermal, meaning its temperature is uniform throughout its entire volume. This allows the system to be solved as a network of interconnected thermal resistances and capacitances, analogous to an electrical circuit, where temperature equates to voltage and heat flow to current flow. [6]

Figure 4.2 shows a simplified thermal system and its equivalent nodal network representation:

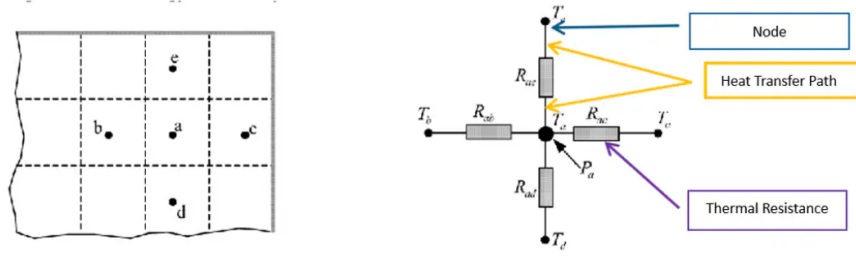


Figure 4.2: Simplified heat transfer system and its corresponding thermal network [8]

The behaviour of the network can be described as a system of differential equations whose coefficients are grouped values of the nodes. The mathematical expression for the nodal energy balance equation is:

$$C_i \frac{dT_i}{dt} = Q_i - \sum_{j=1}^n GL_{ij}(T_i - T_j) - \sum_{j=1}^n GR_{ij}(T_i^4 - T_j^4) \quad (4.4)$$

where:

- C_i [J/K]: thermal capacitance of node i ($m_i c_{p,i}$); stores/releases heat as T_i changes.
- T_i [K]: temperature of node i .
- t [s]: time.
- GL_{ij} [W/K]: conductive conductance between nodes i and j . The value is zero if there is not a conductive path.
- GR_{ij} [W/K⁴]: radiative coupling factor between i and j in the non-linear T^4 form; it embeds σ , view factors and emissivities for diffuse-gray surfaces.
- Q_i [W]: net heat source applied to node i (internal dissipation, heaters, and absorbed external fluxes as modelled in the network).
- $\sum_{j=1}^n$: sum over all nodes j , which are thermally connected to i . The terms $GL_{ij}(T_i - T_j)$ and $GR_{ij}(T_i^4 - T_j^4)$ contribute as inflow when the expression in parentheses is negative.

The discrete approximation simplifies the numerical solution of the thermal problem by reducing a continuous system, governed by differential equations, to a finite set of equations defined over a limited number of nodes. In steady-state conditions, this system becomes algebraic and can be solved using standard numerical techniques, whereas in transient analysis it remains a system of differential equations [6].

Once the energy balance is defined for each node, the complete model is obtained by assembling all the equations into a coupled system. This system describes the temporal evolution of the nodal temperatures, taking into account both internal heat sources and the thermal interactions between nodes. [6]

In this framework, thermal systems are represented as networks of nodes connected through thermal conductances. These conductances account for the main heat transfer mechanisms, such as conduction through structural elements and radiation exchange between surfaces. Each node is characterised by its thermal capacitance and exchanges heat with the rest of the system, enabling the prediction of both steady-state and transient behaviour [2,6].

The accuracy of this method depends on the discretisation of the system. A reduced number of nodes may not capture the relevant temperature gradients, while an excessive number increases the computational cost without providing significant accuracy in the results. Therefore, the definition of the nodes represents a compromise between accuracy and efficiency. [7]

Therefore, due to its balance between accuracy and computational cost, this method is widely used in spacecraft thermal modelling and forms the basis of tools such as ESATAN-TMS. [7]

4.5. Software and Advanced Methods for Thermal Simulation

4.6. Analytical Capabilities of ESATAN-TMS

5. Development of the Thermal Model in ESATAN-TMS and Data Export/Analysis

5.1. Subsystem Classification

5.2. Materials and Properties

5.2.1. Thermo-optical properties

5.3. Steady-state radiative case

5.4. Boundary Conditions and Loads Definition

5.5. Model Verification: Group-Level Energy Balances

5.6. Data provided by the TMD

5.6.1. Use in the Python post-processing

6. Design and Implementation of the Post-Processing Code

6.1. Code Structure and Architecture

6.1.1. Motivation for a Modular Architecture

6.1.2. Files overview

6.2. Software Stack and Dependencies

6.3. Filtering and Grouping Methods

6.4. Figure Conventions and Representation Criteria

6.5. Scope of Examples and Cryogenic-Flow Validation

7. Results and Visualisations from the Post-Processing Workflow

7.1. Filtering Results

7.2. Sankey Diagram

7.3. Network Graph Views (NX)

7.4. Stacked-Bar Balances

7.5. Cryogenic-Flow Views (ARIEL)

8. Conclusions and Future Work

8.1. Conclusions

8.2. Future Work

References

- [1] D. Gilmore (Ed.), *Spacecraft Thermal Control Handbook, Volume I: Fundamental Technologies*, 2nd Edition, The Aerospace Press, El Segundo, CA, 2002.
- [2] J. Meseguer, I. Pérez-Grande, A.Sanz-Andrés, *Spacecraft Thermal Control*, 1st Edition, Woodhead Publishing Limited, Cambridge, UK, 2012.
- [3] I. G. de Control Térmico Espacial, Descripción y objetivos del TCS: Control térmico espacial, diapositivas de la asignatura Transferencia de calor y control térmico, material docente de la Universidad Politécnica de Madrid[consultado el 6 de marzo de 2026] (2026).
- [4] Y. A. Çengel, A. J. Ghajar, *Heat and Mass Transfer: Fundamentals and Applications*, 6th Edition, McGraw-Hill, New York, NY, 2020.
- [5] I. G. de Control Térmico Espacial, Conducción *i*: Thermal conductivity and heat equation unidimensional, diapositivas de la asignatura Transferencia de calor y control térmico, material docente de la Universidad Politécnica de Madrid[consultado el 6 de marzo de 2026] (2026).
- [6] I. E. U. Ltd, *ESATAN-TMS Thermal Engineering Manual*.
- [7] E. S. A. (ESA), *ECSS-E-HB-31-03A: Thermal Analysis Handbook* (2016).
- [8] [Understanding the lumped-parameter thermal network method](#) [online].
- [9] A. B. Gil, D. González-Bárcena, *Principios de Transferencia de Calor*, primera edición Edition, Editorial Garceta, Universidad Politécnica de Madrid, Escuela Técnica Superior de Ingeniería Aeronáutica y del Espacio, 2026.
- [10] I. G. de Control Térmico Espacial, Conducción *ii*: heat equation and thermal resistance, diapositivas de la asignatura Transferencia de calor y control térmico, material docente de la Universidad Politécnica de Madrid[consultado el 6 de marzo de 2026] (2026).
- [11] I. G. de Control Térmico Espacial, Conducción *iii*: thermal contact resistance and thermal straps, diapositivas de la asignatura Transferencia de calor y control térmico, material docente de la Universidad Politécnica de Madrid[consultado el 6 de marzo de 2026] (2026).
- [12] I. G. de Control Térmico Espacial, Convección, diapositivas de la asignatura Transferencia de calor y control térmico, material docente de la Universidad Politécnica de Madrid[consultado el 6 de marzo de 2026] (2026).
- [13] [Aerospace tech, control térmico espacial en vehículos espaciales](#) [online].

Título del anexo

Aquí puedes meter la información que no sea imprescindible en el cuerpo del trabajo pero si que interese que esté en el documento.

